Point-Continuous and Central Valuations: Countable Spaces and the Sorgenfrey–Smyth dcpo

Yuxu Chen[a]

[a]*School of Mathematics, Sichuan University, Chengdu, China*

Abstract

We prove that every continuous valuation on a space whose open sets are saturations of at most countable subsets is point-continuous. In particular, this holds for every countable space. We also prove that the Smyth dcpo of the Sorgenfrey line is a Fubini object, meaning that all its continuous valuations are central. Finally, we prove that if the Scott-open set lattice of a dcpo P is a Fubini object, then P is a Fubini object. Combined with known co-consonance results, this yields that every dcpo with second-countable Scott topology is a Fubini object.

Keywords: continuous valuation, point-continuous valuation, central valuation, countable space, Sorgenfrey line, Fubini theorem
2020 MSC: 06B35, 28C15, 54H05, 68Q55

1. Introduction

Continuous valuations provide a domain-theoretic model of probabilistic choice. They assign nonnegative masses to open sets, satisfy modularity, and preserve directed unions. The valuation-based probabilistic powerdomain is developed in [1]; subprobability valuations form a strong monad on directed-complete partial orders (dcpos) and Scott-continuous maps [2, Sections 4.2–4.3]. The Fubini problem asks whether two continuous valuations always commute when integrated against a nonnegative Scott-continuous function on a product dcpo [3–5]. The Scott topology of the ordered product may be strictly finer than the product of the Scott topologies. Consequently, ordinary topological-product Fubini theorems do not settle this problem [3].

For a topological space X, let $\mathcal{O}(X)$ be its open-set lattice and $\mathbf{V}(X)$ its continuous valuations with values in $[0,\infty]$. A valuation is *point-continuous* if, whenever $0 \le r < \mu(U)$, some finite $F \subseteq U$ satisfies $\mu(V) > r$ for every open neighbourhood V of F [5, Section 2.1]. We write $\mathbf{P}(X)$ for this class. For a dcpo P, let ΣP be its Scott space and $\sigma(P)$ its Scott-open set lattice, and abbreviate $\mathbf{V}(P) = \mathbf{V}(\Sigma P)$ and $\mathbf{P}(P) = \mathbf{P}(\Sigma P)$. A valuation on P is *central* if it commutes with every continuous valuation on every dcpo; write $\mathbf{Z}(P)$ for this class. Point-continuous valuations are central: this follows from the Fubini theorem for point-continuous subprobability valuations [3, Theorem 21], together with the normalization and approximation arguments recalled below. We call P a *Fubini object* when $\mathbf{V}(P) = \mathbf{Z}(P)$.

Email address: chenyuxu@scu.edu.cn (Yuxu Chen)

Our first result is that every continuous valuation on a countable topological space is point-continuous. More generally, this holds when every open set is the saturation of an at most countable subset. The proof shows that the neighbourhood capacity of a countable set is the supremum of the capacities of its finite initial segments. Modularity controls the masses of finite unions of almost-minimal neighbourhoods. For a countable dcpo P, it follows that $\mathbf{V}(P) = \mathbf{P}(P) = \mathbf{Z}(P)$.

Our second result concerns $D = \mathcal{Q}(\mathbb{R}_\ell)$, the Smyth dcpo of the Sorgenfrey line. Goubault-Larrecq and Jia constructed a continuous valuation on D which is not point-continuous [5, Theorem 4.21] and suggested its possible use in a counterexample to general Fubini equations [5, Section 5]. We prove instead that every continuous valuation on D is central; consequently, $\mathbf{P}(D) \subsetneq \mathbf{Z}(D) = \mathbf{V}(D)$. The proof lifts finite valuations to the full Vietoris topology and establishes the canonical incidence identity through a Scott-continuous Fatou inequality, finite measurable partitions, and the Radon–Nikodym theorem. Appendix A proves more generally that every continuous valuation on a subspace of a countable product of Sorgenfrey lines has a central pushforward along every continuous map into the Scott space of a dcpo. Its proof of the incidence identity uses discrete threshold minorants and one-sided approximation; the application to D retains the same full Vietoris lift.

The final section records consequences rather than additional independent constructions. A short incidence argument transfers the Fubini property from $\sigma(P)$ to P. Together with de Brecht and Kawai's co-consonance theorem [6, Proposition 7.2], it yields the Fubini property for every dcpo with second-countable Scott topology.

2. Preliminaries

A nonempty subset A of a poset is *directed* if every finite subset of A has an upper bound in A. A *dcpo* is a poset in which every directed subset has a supremum. A subset U of a dcpo P is *Scott open* if it is upper and, whenever $A \subseteq P$ is directed with $\sup A \in U$, one has $A \cap U \neq \varnothing$. We write ΣP for its Scott space and $\sigma(P)$ for its Scott-open set lattice, ordered by inclusion. A map between dcpos is *Scott-continuous* if it is monotone and preserves directed suprema, equivalently if it is continuous between their Scott spaces [7, 8].

For a topological space X, $\mathcal{O}(X)$ is its complete lattice of open subsets, ordered by inclusion, and $\Sigma\mathcal{O}(X)$ is the Scott space of that lattice. A space is *countably based*, or *second-countable*, if its topology has a countable base; it is *first-countable* if each point has a countable neighbourhood base. For a set A, $\mathcal{P}(A)$ denotes its power set ordered by inclusion.

A subset is *compact* if every open cover has a finite subcover; no separation axiom is implicit. Its saturation is $\mathrm{sat}(A) = \bigcap\{V \in \mathcal{O}(X) : A \subseteq V\}$, and it is *saturated* when $A = \mathrm{sat}(A)$. For $A \subseteq P$, write $\uparrow A = \{x \in P : (\exists a \in A)\ a \leq x\}$. In a Scott space saturation equals upper closure [8]. A subset $A \subseteq B \subseteq P$ is *order-coinitial in* B if every $b \in B$ lies above some $a \in A$.

2.1. Continuous valuations and integration

Let $\overline{\mathbb{R}}_+ = [0, \infty]$ with its usual order and Scott topology. A *continuous valuation* on X is a strict, monotone map $\nu : \mathcal{O}(X) \to \overline{\mathbb{R}}_+$ satisfying modularity, $\nu(U) + \nu(V) = \nu(U \cup V) + \nu(U \cap V)$, and directed continuity, $\nu(\bigcup_i U_i) = \sup_i \nu(U_i)$. Strictness means $\nu(\varnothing) = 0$. It is *finite*, or *bounded*, if $\nu(X) < \infty$, a *subprobability valuation* if $\nu(X) \leq 1$, and a *probability valuation* if $\nu(X) = 1$. Write

$\mathbf{V}(X)$ and $\mathbf{V}_{\leq 1}(X)$ for continuous and subprobability valuations, respectively. They are dcpos under the pointwise order, with directed suprema computed pointwise. Addition and multiplication by finite nonnegative scalars are pointwise, with $0 \cdot \infty = 0$. For a dcpo P, abbreviate $\mathbf{V}(P) = \mathbf{V}(\Sigma P)$ and $\mathbf{V}_{\leq 1}(P) = \mathbf{V}_{\leq 1}(\Sigma P)$ [2, 8].

A function $h : X \to \overline{\mathbb{R}}_+$ is *lower semicontinuous* if $\{x : h(x) > r\}$ is open for every finite $r \geq 0$. Its Choquet integral is

$$\int_X h \, d\nu = \int_0^\infty \nu(h^{-1}((t, \infty])) \, dt,$$

where the right-hand integral is the ordinary Lebesgue integral. The Choquet integral is additive, positively homogeneous, and preserves directed suprema in the integrand [9, Theorem 3.5]. It also preserves directed suprema in the valuation: this holds for lower-step functions by pointwise computation and hence for all nonnegative lower-semicontinuous functions by lower-step approximation. For a continuous $g : X \to Y$, write $g_*\nu$ for the valuation defined by $(g_*\nu)(V) = \nu(g^{-1}(V))$. The change-of-variables formula is

$$\int_Y h \, d(g_*\nu) = \int_X (h \circ g) \, d\nu. \tag{2.1}$$

Indeed, for every finite $t \geq 0$, $g^{-1}(h^{-1}((t, \infty])) = (h \circ g)^{-1}((t, \infty])$, so the identity follows directly from the layer-cake definition of the Choquet integral.

For $A \subseteq X$, let $\mathbf{1}_A$ denote its characteristic function. The Dirac valuation is $\delta_x(U) = \mathbf{1}_U(x)$; a *simple valuation* is a finite sum $\sum_{i=1}^m a_i \delta_{x_i}$ with $a_i \geq 0$ finite. Then $\int_X h \, d(\sum_i a_i \delta_{x_i}) = \sum_i a_i h(x_i)$.

2.2. *Point-continuity, centrality, and Fubini objects*

The *point topology* on $\mathcal{O}(X)$ is generated by $M_x = \{U \in \mathcal{O}(X) : x \in U\}$, $x \in X$. A continuous valuation is *point-continuous* if it is continuous from this topology to $\overline{\mathbb{R}}_+$. Write $\mathbf{P}(X)$ for these valuations and $\mathbf{P}(P) = \mathbf{P}(\Sigma P)$ for a dcpo P.

Valuations $\mu \in \mathbf{V}(P)$ and $\nu \in \mathbf{V}(Q)$ on dcpos *commute* if every Scott-continuous $h : P \times Q \to \overline{\mathbb{R}}_+$ satisfies

$$\int_P \left(\int_Q h(x, y) \, d\nu(y) \right) d\mu(x) = \int_Q \left(\int_P h(x, y) \, d\mu(x) \right) d\nu(y). \tag{2.2}$$

The inner-integral functions are Scott-continuous by directed continuity of integration. A valuation is *central* if it commutes with every continuous valuation on every dcpo; write $\mathbf{Z}(P)$ for this class. We use the *extended* convention: neither mass nor integrand is required to be bounded. In this convention, central valuations are closed under multiplication by finite nonnegative scalars, finite sums and directed suprema. These closure properties follow from linearity and directed continuity of the two iterated integrals. In particular, for a directed family $(\mu_i)_i$ and $\mu = \sup_i \mu_i$, each side of (2.2) with μ is the supremum of its corresponding sides with μ_i. Thus equality for every μ_i passes to μ.

The usual probabilistic convention restricts the two valuations to subprobabilities and the integrand to $[0, 1]$. Let $\mathbf{Z}_{\leq 1}(P)$ denote the subprobability valuations central in this sense [4]. We recall Heckmann's approximation theorem, then check the relation between the two conventions.

Theorem 2.1. *[10, Theorem 4.2] On a T_0 space, every continuous valuation is a directed supremum of bounded continuous valuations, and every point-continuous valuation is a directed supremum of bounded point-continuous valuations.*

Proposition 2.2. *For every dcpo P, $\mathbf{Z}_{\leq 1}(P) = \mathbf{Z}(P) \cap \mathbf{V}_{\leq 1}(P)$. The following conditions are equivalent: all subprobability valuations on P are central in the subprobability sense; all finite continuous valuations on P are central in the extended sense; all continuous valuations on P are central in the extended sense.*

Proof. Let $\mu \in \mathbf{Z}_{\leq 1}(P)$, let $\nu \in \mathbf{V}(Q)$ be finite, and take a Scott-continuous $h : P \times Q \to \overline{\mathbb{R}}_+$. Put $c = \max\{1, \nu(Q)\}$. Apply subprobability centrality to $c^{-1}\nu$ and $n^{-1}(h \wedge n)$, then multiply by cn. This gives (2.2) for $h \wedge n$. Since these functions increase to h, directed continuity gives the equality for h.

For arbitrary ν, write $\nu = \sup_i \nu_i$ with $(\nu_i)_i$ directed and finite, using Theorem 2.1. Each iterated integral preserves this directed supremum, by applying directed continuity to its inner and outer integral. Equality for the ν_i therefore yields equality for ν. Hence $\mathbf{Z}_{\leq 1}(P) \subseteq \mathbf{Z}(P)$, while the reverse inclusion after intersection with $\mathbf{V}_{\leq 1}(P)$ is immediate.

If every subprobability valuation on P is central, divide a finite μ by $\max\{1, \mu(P)\}$ and apply the first assertion and scalar closure. If every finite valuation is central, Heckmann's approximation theorem and closure under directed suprema show that every valuation is central. The remaining implication follows by restriction to subprobabilities. □

Proposition 2.3. *[5, Section 2.1] A continuous valuation ν on X is point-continuous if and only if, for every $U \in \mathcal{O}(X)$ and $0 \leq r < \nu(U)$, some finite $F \subseteq U$ satisfies $\nu(V) > r$ whenever $F \subseteq V \in \mathcal{O}(X)$.*

Theorem 2.4. *[3, Theorem 21] Let P, Q be dcpos, $\mu \in \mathbf{P}(P) \cap \mathbf{V}_{\leq 1}(P)$ and $\nu \in \mathbf{V}_{\leq 1}(Q)$. For every Scott-open $W \subseteq P \times Q$, $\int_P \nu(W_x)\, d\mu(x) = \int_Q \mu(W^y)\, d\nu(y)$, where $W_x = \{y : (x, y) \in W\}$ and $W^y = \{x : (x, y) \in W\}$.*

The theorem is stated for characteristic functions of Scott-open sets. Dyadic lower-step approximation, followed by directed continuity of integration, extends it to nonnegative Scott-continuous integrands. Together with Theorem 2.1 and Proposition 2.2, this gives the following inclusion in the extended convention.

Corollary 2.5. *For every dcpo P, $\mathbf{P}(P) \subseteq \mathbf{Z}(P) \subseteq \mathbf{V}(P)$.*

Proof. For a finite point-continuous μ, scale to a subprobability valuation. If $h : P \times Q \to [0, 1]$ is Scott-continuous, then

$$h_m = 2^{-m} \sum_{k=1}^{2^m - 1} \mathbf{1}_{\{h > k2^{-m}\}}$$

increases pointwise to h. The preceding theorem, linearity, and directed continuity therefore give subprobability centrality for every such h. By Proposition 2.2, μ is central in the extended sense. For arbitrary $\mu \in \mathbf{P}(P)$, use the point-continuous part of Theorem 2.1 and closure of central valuations under directed suprema. □

Definition 2.6. A dcpo P is a *Fubini object* if $\mathbf{V}(P) = \mathbf{Z}(P)$. Write Fub for this class.

The assertion $\mu \in \mathbf{Z}(P)$ concerns one valuation, whereas $P \in \mathsf{Fub}$ quantifies over all valuations on P. In the latter case the partner dcpo is still arbitrary. By Proposition 2.2, $P \in \mathsf{Fub}$ is equivalent to $\mathbf{V}_{\leq 1}(P) = \mathbf{Z}_{\leq 1}(P)$.

2.3. Canonical incidence

For a dcpo P, let $E_P = \{(x, U) \in P \times \sigma(P) : x \in U\}$ and $M_x = \{U \in \sigma(P) : x \in U\}$. We use the standard Scott-continuity of evaluation.

Lemma 2.7. *The relation E_P is Scott open in $P \times \sigma(P)$, and $n_P : P \to \sigma(\sigma(P))$, $n_P(x) = M_x$, is Scott-continuous.*

Proof. For each $x \in P$, the set M_x is Scott open, since membership of x in a directed union is witnessed by one member. The relation E_P is upper. Let $((x_i, U_i))_{i\in I}$ be directed in $P \times \sigma(P)$, with supremum (x, U), and suppose that $x \in U$. Since $U = \bigcup_i U_i$, some i satisfies $x \in U_i$. Scott openness of U_i and $x = \sup_j x_j$ give an index j with $x_j \in U_i$. Choose k above both i and j. Then $x_j \leq x_k$ and $U_i \subseteq U_k$, so $x_k \in U_k$. Hence E_P is Scott open.

The map n_P is monotone. If $(x_i)_i$ is directed with supremum x, then, for every $U \in \sigma(P)$,

$$U \in M_x \iff x \in U \iff \exists i.\, x_i \in U \iff U \in \bigcup_i M_{x_i}.$$

Thus $M_x = \bigcup_i M_{x_i}$, which is the directed supremum in $\sigma(\sigma(P))$, and n_P is Scott-continuous. □

The following reduction is an elementary consequence of evaluation and change of variables.

Proposition 2.8. *A valuation $\mu \in \mathbf{V}(P)$ is central if and only if every $\xi \in \mathbf{V}(\sigma(P))$ satisfies*

$$\int_{\sigma(P)} \mu(U)\, d\xi(U) = \int_P \xi(M_x)\, d\mu(x). \tag{2.3}$$

Proof. Necessity follows by applying centrality to $\mathbf{1}_{E_P}$. Conversely, take $\nu \in \mathbf{V}(Q)$ and Scott-open $W \subseteq P \times Q$. The section map $s_W : Q \to \sigma(P)$, $s_W(y) = W^y$, is monotone and satisfies $s_W(\sup A) = \bigcup_{y\in A} s_W(y)$ for each directed $A \subseteq Q$, hence is Scott-continuous. Define $\xi = (s_W)_*\nu$; explicitly, $\xi(\mathcal{H}) = \nu(\{y : W^y \in \mathcal{H}\})$ for $\mathcal{H} \in \sigma(\sigma(P))$. Since $s_W^{-1}(M_x) = W_x$, equation (2.3) and change of variables give $\int_P \nu(W_x)\, d\mu(x) = \int_Q \mu(W^y)\, d\nu(y)$.

For general Scott-continuous $h : P \times Q \to \overline{\mathbb{R}}_+$, use the lower-step functions

$$h_m = 2^{-m} \sum_{k=1}^{m2^m} \mathbf{1}_{\{h>k2^{-m}\}}.$$

The dyadic grids are nested and the truncation levels increase, so $h_m \leq h_{m+1}$. If $h(p, q) < \infty$, then for all sufficiently large m the error $h(p, q) - h_m(p, q)$ is at most 2^{-m}; if $h(p, q) = \infty$, then $h_m(p, q) = m$. Hence $\sup_m h_m = h$. Linearity gives (2.2) for each h_m, and directed continuity gives it for h. □

Corollary 2.9. *A continuous valuation $\mu \in \mathbf{V}(P)$ is central if and only if (2.3) holds for every finite continuous valuation ξ on $\sigma(P)$, equivalently, for every $\xi \in \mathbf{V}_{\leq 1}(\sigma(P))$.*

Proof. Both sides of (2.3) are homogeneous in ξ and preserve its directed suprema. Thus subprobability tests imply finite tests by scaling, and finite tests imply all tests by Theorem 2.1. Apply Proposition 2.8. □

3. Countable spaces and point-continuity

Let X be a topological space and let $\nu \in \mathbf{V}(X)$. For $A \subseteq X$, define the neighbourhood capacity

$$\kappa_\nu(A) = \inf\{\nu(V) : A \subseteq V \in O(X)\}. \tag{3.1}$$

Lemma 3.1. *Let X be a topological space, let $\nu \in \mathbf{V}(X)$, and let $F_1 \subseteq F_2 \subseteq \cdots$ be finite subsets of X with $A = \bigcup_n F_n$. Then*

$$\kappa_\nu(A) = \sup_{n \geq 1} \kappa_\nu(F_n). \tag{3.2}$$

Proof. Put $c_n = \kappa_\nu(F_n)$ and $c = \sup_n c_n$. The inequality $c \leq \kappa_\nu(A)$ is immediate. Suppose that $c < \kappa_\nu(A)$. Then $c < \infty$. Choose $\delta > 0$ with $c + \delta < \kappa_\nu(A)$ and positive numbers ε_n with $\sum_n \varepsilon_n < \delta$. For each n, choose an open set $V_n \supseteq F_n$ such that

$$\nu(V_n) < c_n + \varepsilon_n. \tag{3.3}$$

Set $W_n = V_1 \cup \cdots \cup V_n$. Each V_n has finite mass, so $\nu(W_n) \leq \sum_{k=1}^n \nu(V_k) < \infty$. All subsequent subtractions are therefore between finite numbers. We prove by induction that

$$\nu(W_n) < c_n + \sum_{k=1}^{n} \varepsilon_k. \tag{3.4}$$

The case $n = 1$ is (3.3). For $n \geq 2$, the inclusion $F_{n-1} \subseteq W_{n-1} \cap V_n$ gives $\nu(W_{n-1} \cap V_n) \geq c_{n-1}$. Modularity and the induction hypothesis yield

$$\begin{aligned} \nu(W_n) &= \nu(W_{n-1}) + \nu(V_n) - \nu(W_{n-1} \cap V_n) \\ &< c_{n-1} + \sum_{k=1}^{n-1} \varepsilon_k + c_n + \varepsilon_n - c_{n-1}, \end{aligned}$$

which is (3.4).

Now $W = \bigcup_n W_n$ is an open neighbourhood of A. Directed continuity gives

$$\kappa_\nu(A) \leq \nu(W) = \sup_n \nu(W_n) \leq c + \sum_n \varepsilon_n < c + \delta < \kappa_\nu(A),$$

a contradiction. □

Theorem 3.2. *Let X be a topological space such that every open set is the saturation of an at most countable subset. Then $\mathbf{V}(X) = \mathbf{P}(X)$. In particular, every continuous valuation on a countable topological space is point-continuous.*

Proof. Let $\nu \in \mathbf{V}(X)$, let $U \in \mathcal{O}(X)$, and let $0 \le r < \nu(U)$. Choose an at most countable $A \subseteq U$ with $U = \mathrm{sat}(A)$. The sets A and U have the same open neighbourhoods, and U itself is one of them. Hence $\kappa_\nu(A) = \nu(U)$.

Choose finite sets $F_1 \subseteq F_2 \subseteq \cdots$ with $A = \bigcup_n F_n$. By Lemma 3.1,

$$\nu(U) = \sup_{n \ge 1} \kappa_\nu(F_n).$$

Thus $\kappa_\nu(F_n) > r$ for some n. Every open V containing F_n satisfies $\nu(V) \ge \kappa_\nu(F_n) > r$. Heckmann's finite-point criterion, Proposition 2.3, gives point-continuity. For countable X, take $A = U$. □

Applying Theorem 3.2 and Corollary 2.5 to ΣP gives the following consequence.

Corollary 3.3. *Let P be a dcpo such that every Scott-open subset has an at most countable order-coinitial subset. Then $\mathbf{V}(P) = \mathbf{P}(P) = \mathbf{Z}(P)$. In particular, every countable dcpo is a Fubini object.*

4. The Sorgenfrey–Smyth dcpo

Let $\mathbb{R}_\ell$ be the Sorgenfrey line, whose topology has the intervals $[a, b)$, $a < b$, as a basis, and let $D = \mathcal{Q}(\mathbb{R}_\ell)$ be the set of its nonempty compact subsets, ordered by reverse inclusion. D is a dcpo and its Scott topology is the upper Vietoris topology, with basis

$$\Box U = \{K \in D : K \subseteq U\}, \qquad U \in \mathcal{O}(\mathbb{R}_\ell),$$

by [11, Theorem 5.8 and Example 5.18].

We will prove that every continuous valuation on D is central. The proof first establishes an incidence identity for finite measures on first-countable spaces with a countable family of inner approximating opens. It then lifts a finite valuation on D to the full Vietoris topology and verifies the approximation condition using countably many extremal maps. Throughout this section, a finite Borel measure is called *τ-smooth* when its restriction to open sets is a continuous valuation. For such a measure, the Lebesgue and Choquet integrals of nonnegative lower-semicontinuous functions agree by [9, Proposition 4.3].

4.1. A measure-theoretic incidence criterion

We use the following Fatou-type inequality for Scott-continuous functions. The Scott continuity of the lower bound is essential to the argument since an arbitrary pointwise infimum of Scott-continuous functions need not be Scott-continuous.

Lemma 4.1. *Let P be a dcpo, let $\xi \in \mathbf{V}(P)$, and let $h, h_1, h_2, \ldots : P \to \overline{\mathbb{R}}_+$ be Scott-continuous. If*

$$h(p) \le \liminf_{n \to \infty} h_n(p) \qquad (p \in P),$$

then

$$\int_P h \, d\xi \le \liminf_{n \to \infty} \int_P h_n \, d\xi. \tag{4.1}$$

Proof. For $N \geq 1$, define

$$k_N(p) = \min\left\{h(p), \inf_{n\geq N} h_n(p)\right\}.$$

We first prove that k_N is Scott-continuous. It is monotone. Let $A \subseteq P$ be directed, let $p = \sup A$, and take $0 \leq r < k_N(p)$. Choose $q \in [0,\infty)$ with $r < q < k_N(p)$. Since $h(p) > q$, there is $a_0 \in A$ with $h(a_0) > q$. The assumed lower-limit inequality at a_0 gives $n_0 \geq N$ such that

$$h_n(a_0) > q \qquad (n \geq n_0).$$

For each of the finitely many indices $N \leq n < n_0$, Scott continuity and $h_n(p) > q$ give an element of A where $h_n > q$. Choose a common upper bound $a \in A$ of these elements and a_0. Monotonicity then gives $h(a) > q$ and $h_n(a) > q$ for every $n \geq N$. Hence $k_N(a) \geq q > r$. This proves Scott continuity.

The functions k_N increase, and their pointwise supremum is h. Moreover, $k_N \leq h_n$ whenever $n \geq N$. Directed continuity and monotonicity of integration yield

$$\int_P h\, d\xi = \sup_N \int_P k_N\, d\xi \leq \sup_N \inf_{n\geq N} \int_P h_n\, d\xi,$$

which is (4.1). □

We also record a consequence of finiteness that will be used below. If m is a finite τ-smooth Borel measure on a space X and $B \subseteq X$ is Borel, then

$$\nu_B(U) = m(B \cap U), \qquad U \in \mathcal{O}(X), \tag{4.2}$$

is a continuous valuation. Indeed, for a directed family $(U_i)_i$ with union U,

$$0 \leq m(B \cap U) - m(B \cap U_i) \leq m(U) - m(U_i),$$

and the right-hand side has infimum zero. The other valuation axioms follow from the measure axioms. In particular, $m_B(E) = m(B \cap E)$ is a finite τ-smooth Borel measure on X.

Proposition 4.2. *Let X be a first-countable space and let m be a finite τ-smooth Borel measure on X. Suppose that a countable family $\mathcal{C} \subseteq \mathcal{O}(X)$ satisfies*

$$m(U) = \sup\{m(C) : C \in \mathcal{C},\ C \subseteq U\} \qquad (U \in \mathcal{O}(X)), \tag{4.3}$$

where the supremum of the empty set is 0. Put $L = \mathcal{O}(X)$ and $M_x = \{U \in L : x \in U\}$. Then, for every finite $\eta \in \mathbf{V}(L)$, the function $x \mapsto \eta(M_x)$ is lower semicontinuous and

$$\int_L m(U)\, d\eta(U) = \int_X \eta(M_x)\, dm(x). \tag{4.4}$$

Proof. Fix η, put $b = \eta(L)$, and write $\nu(U) = m(U)$. For $\rho \in \mathbf{V}(X)$, set

$$J(\rho) = \int_L \rho(U)\, d\eta(U), \qquad g(x) = \eta(M_x) = J(\delta_x).$$

Each M_x is Scott open, and $0 \le g \le b$. If $x_n \to x$, then

$$\delta_x(U) \le \liminf_n \delta_{x_n}(U) \qquad (U \in L).$$

Applying Lemma 4.1 on the dcpo L gives $g(x) \le \liminf_n g(x_n)$. Thus g is sequentially lower semicontinuous, and first-countability makes it lower semicontinuous. In particular, g is Borel measurable. If $m(X) = 0$ or $b = 0$, the desired equality is immediate. We henceforth exclude these cases.

The inequality $J(\nu) \le \int_X g\, dm$. Add $\varnothing$ to C if necessary and enumerate it as $C_1, C_2, \ldots$, repeating members if the family is finite. For each n, take the finite Borel partition $\mathcal{A}_n$ generated by $C_1, \ldots, C_n$ and the sets

$$\{x : j2^{-n} \le g(x) < (j+1)2^{-n}\}, \qquad 0 \le j \le \lfloor 2^n b \rfloor.$$

These sets cover X, since $0 \le g \le b$. On each part, g has oscillation at most 2^{-n}. Choose $x_A \in A$ for every positive-mass part $A \in \mathcal{A}_n$, and put

$$\nu_n = \sum_{\substack{A \in \mathcal{A}_n \\ m(A) > 0}} m(A)\delta_{x_A}.$$

These are simple valuations, and

$$\nu_n(C_k) = m(C_k) \quad (k \le n), \qquad \left| J(\nu_n) - \int_X g\, dm \right| \le 2^{-n} m(X). \tag{4.5}$$

The first identity holds because each C_k, $k \le n$, is a union of partition parts. For the second, linearity gives $J(\nu_n) = \sum_A m(A) g(x_A)$, and the oscillation estimate bounds the error on each part by $2^{-n} m(A)$; zero-mass parts contribute nothing.

If $C_k \subseteq U$, then $\nu_n(U) \ge m(C_k)$ for all $n \ge k$. Taking the supremum over these C_k and using (4.3) yields

$$\nu(U) \le \liminf_n \nu_n(U) \qquad (U \in L).$$

Since ν and the ν_n are Scott-continuous functions on L, Lemma 4.1 and (4.5) give

$$J(\nu) \le \liminf_n J(\nu_n) = \int_X g\, dm.$$

Neither directedness of $(\nu_n)_n$ nor the inequalities $\nu_n \le \nu$ are required.

The inequality $\int_X g\, dm \le J(\nu)$. For every Borel set $B \subseteq X$, define

$$\Lambda(B) = \int_L m(B \cap U)\, d\eta(U) = J(\nu_B).$$

This integral is defined by (4.2). For pairwise disjoint Borel sets B_j, the functions $U \mapsto \sum_{j=1}^n m(B_j \cap U)$ are Scott-continuous and increase to $U \mapsto m((\bigcup_j B_j) \cap U)$. Linearity and monotone convergence of valuation integrals show that

$$\Lambda\left(\bigcup_j B_j\right) = \sum_j \Lambda(B_j).$$

Thus Λ is a finite Borel measure, and

$$0 \le \Lambda(B) \le b\, m(B) \qquad (B \subseteq X \text{ Borel}).$$

Since $m(X) < \infty$ and $\Lambda \le bm$, the measure Λ is absolutely continuous with respect to m. In particular, for every $\varepsilon > 0$, $m(B) < \varepsilon/b$ implies $\Lambda(B) < \varepsilon$, so the absolute-continuity hypothesis of [12, 232G] is satisfied. This finite-measure Radon–Nikodym theorem, together with the everywhere-defined measurable representative supplied by [12, 232H(e)], gives a Borel measurable function $f : X \to \mathbb{R}$ such that

$$\Lambda(B) = \int_B f\, dm \qquad (B \subseteq X \text{ Borel}); \tag{4.6}$$

the domination $\Lambda \le bm$ implies $0 \le f \le b$ almost everywhere. Replacing f by $\min\{b, \max\{0, f\}\}$ gives a Borel representative satisfying $0 \le f \le b$ everywhere.

We claim that $g \le f$ almost everywhere. Otherwise there is a rational $a \ge 0$ such that the Borel set

$$B = \{x : f(x) < a < g(x)\}$$

has positive m-measure. The measure m_B is τ-smooth by (4.2). For a finite Borel measure ρ, write $\operatorname{supp}(\rho) = X \setminus \bigcup\{V \in \mathcal{O}(X) : \rho(V) = 0\}$. This is the intersection of its closed sets of full measure, as in [9, Definition 5.1]. When ρ is τ-smooth, its support has full measure by [9, Proposition 5.2]. Consequently there is $x \in B \cap \operatorname{supp}(m_B)$. By [9, Corollary 5.4], every open neighbourhood V of x has positive m_B-measure, that is, $m(B \cap V) > 0$.

Choose a decreasing open neighbourhood base $(V_n)_n$ at x, and define continuous probability valuations

$$\rho_n(U) = \frac{m(B \cap V_n \cap U)}{m(B \cap V_n)}, \qquad U \in L.$$

If $x \in U$, then $V_n \subseteq U$ eventually, so $\rho_n(U) = 1$ eventually. Hence $\delta_x(U) \le \liminf_n \rho_n(U)$ for every $U \in L$. Another application of Lemma 4.1, followed by (4.6), gives

$$g(x) \le \liminf_n J(\rho_n) = \liminf_n \frac{\displaystyle\int_{B \cap V_n} f\, dm}{m(B \cap V_n)} \le a,$$

contrary to $x \in B$. This proves the claim. Therefore

$$\int_X g\, dm \le \int_X f\, dm = \Lambda(X) = J(\nu),$$

and the two inequalities establish (4.4). □

The Radon–Nikodym argument takes place on the Borel σ-algebra of X. It requires no measure extension of η, and no measurability assertion about membership in the ordinary product $X \times \Sigma\mathcal{O}(X)$.

4.2. Lifting to the full Vietoris topology

Let D_{V} denote the set D with the full Vietoris topology, generated by the sets $\Box U$ and

$$\Diamond U = \{K \in D : K \cap U \neq \varnothing\}, \qquad U \in \mathcal{O}(\mathbb{R}_\ell).$$

The identity $D_{\mathrm{V}} \to \Sigma D$ is continuous. We first state a trace-measure observation that avoids a separate refinement of this topology. A *compact ordered space* is a compact Hausdorff space with a closed partial order; its *upward topology* consists of its open upper sets. For the finite measures used below, Radon inner regularity means that the measure of each Borel set is the supremum of the measures of its compact subsets.

Lemma 4.3. *Let M be a finite τ-smooth Borel measure on a space E, and let $A \subseteq E$. Suppose that every Borel subset of $E \setminus A$ is M-null. Then*

$$m(A \cap B) = M(B) \qquad (B \subseteq E \text{ Borel}) \tag{4.7}$$

defines a finite τ-smooth Borel measure on the subspace A.

Proof. The Borel σ-algebra of A is the trace of that of E. If $A \cap B = A \cap B'$, then the Borel set $B \mathbin{\triangle} B'$ is disjoint from A and is null, proving that (4.7) is well-defined. If the traces $A \cap B_n$ are pairwise disjoint, put $C_n = B_n \setminus \bigcup_{k<n} B_k$. The sets C_n are disjoint, have the same traces as the B_n, and satisfy $M(C_n) = M(B_n)$. Countable additivity of M gives countable additivity of m.

Let $(U_i)_i$ be a directed family of open subsets of A, and choose ambient open sets O_i with $U_i = A \cap O_i$. The sets O_i need not be directed, but their finite unions are. Hence

$$\begin{aligned} m\left(\bigcup_i U_i\right) &= M\left(\bigcup_i O_i\right) \\ &= \sup_{F \text{ finite}} M\left(\bigcup_{i \in F} O_i\right) \\ &= \sup_{F \text{ finite}} m\left(\bigcup_{i \in F} U_i\right) = \sup_i m(U_i), \end{aligned}$$

where the last equality uses the directedness of $(U_i)_i$. This proves τ-smoothness. □

Lemma 4.4. *For every finite $\mu \in \mathbf{V}(D)$, there is a finite τ-smooth Borel measure m on D_{V} such that*

$$m(\mathcal{U}) = \mu(\mathcal{U}) \qquad (\mathcal{U} \in \sigma(D)). \tag{4.8}$$

Proof. The Sorgenfrey line is Hausdorff with a clopen basis, hence Tychonoff. Choose a compact Hausdorff compactification $\mathbb{R}_\ell \subseteq T$, and put $E = \mathcal{Q}(T)$, ordered by reverse inclusion. It is a dcpo, with directed suprema given by intersection. Its Scott topology is the upper Vietoris topology by [11, Proposition 5.3], and its full Vietoris topology is compact Hausdorff by [13, Theorem 4.9.6].

The reverse-inclusion order on E is closed. Indeed, if $H \not\subseteq K$, choose $x \in H \setminus K$ and disjoint open neighbourhoods V of x and W of K in T. Then $\Box W \times \Diamond V$ contains (K, H) and is disjoint from the relation $K' \supseteq H'$.

The full Vietoris topology is generated by the Scott opens and the sets $E \setminus {\uparrow}H$, $H \in E$, because

$$E \setminus {\uparrow}H = \Diamond(T \setminus H),$$

and the complement of an open subset of T is compact; the cases $\Diamond T = E$ and $\Diamond\varnothing = \varnothing$ are harmless. Its open upper sets are exactly the Scott opens. To see the nontrivial inclusion, suppose an upper full-Vietoris-open set G contains the supremum K of a directed family $(K_i)_i$. Choose a basic neighbourhood

$$V \cap \bigcap_{j=1}^{r} (E \setminus {\uparrow}H_j) \subseteq G$$

of K, with V Scott open. Some K_i lies in V. Since $K_i \leq K$ and $K \notin {\uparrow}H_j$, that K_i belongs to every $E \setminus {\uparrow}H_j$, hence to G.

The inclusion $j : D \to E$ is an embedding for both Vietoris topologies. Indeed, if $O \subseteq T$ is open and $U = O \cap \mathbb{R}_\ell$, then

$$j^{-1}(\Box O) = \Box U, \qquad j^{-1}(\Diamond O) = \Diamond U.$$

Conversely, since $\mathbb{R}_\ell$ carries the subspace topology inherited from T, every open $U \subseteq \mathbb{R}_\ell$ has the form $O \cap \mathbb{R}_\ell$ for some open $O \subseteq T$. Thus the subspace topologies induced by j agree with the upper and, respectively, the full Vietoris topologies on D.

Its image $A = j(D)$ is upper. Indeed, if $K \in A$, $H \in E$, and $H \subseteq K$, then H is closed in T, hence closed in the compact subspace $K \subseteq \mathbb{R}_\ell$; therefore H is compact in $\mathbb{R}_\ell$ and belongs to A.

Define $\lambda(V) = \mu(j^{-1}(V))$ for $V \in \sigma(E)$. This is a finite continuous valuation because j is continuous for the upper Vietoris topologies. The preceding paragraphs show that the full Vietoris space, together with the reverse-inclusion order, is a compact ordered space and that its upward topology is exactly $\sigma(E)$. Moreover,

$$\lambda(E) = \mu(D) < \infty.$$

Hence the Radon extension theorem for compact ordered spaces, [14, Theorem 27], applies and yields a unique finite Radon measure M on the full Vietoris space E agreeing with λ on Scott opens. We show that every Borel subset of $E \setminus A$ is M-null. By inner regularity it suffices to consider a compact $C \subseteq E \setminus A$. Closedness of the order implies that

$${\downarrow}C = \{H \in E : (\exists K \in C)\ K \subseteq H\}$$

is compact: it is the projection of a closed subset of the compact space $E \times C$. Its complement is an open upper set and hence Scott open. Since A is upper, it is disjoint from ${\downarrow}C$, and consequently

$$M(E \setminus {\downarrow}C) = \lambda(E \setminus {\downarrow}C) = \mu(D) = M(E).$$

It follows that $M(C) = 0$, proving the claim.

Every Radon measure on a Hausdorff space is τ-smooth; see the discussion following [9, Definition 4.1]. Apply Lemma 4.3 and transport its trace measure on A along the full Vietoris embedding j. The resulting measure on D_{V} is finite and τ-smooth. For a Scott-open $\mathcal{U} \subseteq D$, the upper Vietoris embedding gives $V \in \sigma(E)$ with $j^{-1}(V) = \mathcal{U}$, and then

$$m(\mathcal{U}) = M(V) = \lambda(V) = \mu(\mathcal{U}).$$

This is (4.8). □

4.3. Countable observations and centrality

Let $\mathscr{A}$ be the countable Boolean algebra generated by the clopen rays $(-\infty, q)$, $q \in \mathbb{Q}$. For $C \in \mathscr{A}$, define

$$\ell_C(K) = \min(K \cap C), \qquad u_C(K) = \max(K \cap C)$$

when $K \cap C \neq \varnothing$, and give both maps the value $*$ otherwise. The codomain is the topological sum $\mathbb{R}_\ell \sqcup \{*\}$, with $*$ isolated. Since C is clopen, $K \cap C$ is compact in the Sorgenfrey line. Its image under the identity map to the ordinary real line is compact, and hence has a minimum and a maximum.

Lemma 4.5. *The maps ℓ_C, u_C, $C \in \mathscr{A}$, are continuous and jointly generate the full Vietoris topology. In particular, D_{V} is first-countable.*

Proof. Empty and nonempty intersection with C are both full Vietoris-open conditions. On the nonempty part,

$$\begin{aligned}
\ell_C(K) \geq a &\iff K \subseteq (\mathbb{R}_\ell \setminus C) \cup [a, \infty), \\
\ell_C(K) < b &\iff K \cap C \cap (-\infty, b) \neq \varnothing, \\
u_C(K) \geq a &\iff K \cap C \cap [a, \infty) \neq \varnothing, \\
u_C(K) < b &\iff K \subseteq (\mathbb{R}_\ell \setminus C) \cup (-\infty, b).
\end{aligned}$$

These formulas prove continuity.

The topology generated by $\mathscr{A}$ on $\mathbb{R}$ is coarser than the Sorgenfrey topology and Hausdorff: a rational cut separates any two distinct real numbers. On a Sorgenfrey-compact K, the two subspace topologies coincide by the compact-to-Hausdorff homeomorphism theorem. Thus every finite open cover of K admits a finite refinement on K by members of $\mathscr{A}$. Taking the atoms of the resulting finite Boolean algebra gives a finite partition of $\mathbb{R}$ by members of $\mathscr{A}$ with the same refinement property on K.

For $K \subseteq U$, cover K by finitely many intervals contained in U and choose such a partition $(C_j)_j$. Require every empty fragment to remain empty. For every nonempty fragment $K \cap C_j \subseteq [a_j, b_j) \subseteq U$, require nonemptiness and

$$\ell_{C_j}(K') \geq a_j, \qquad u_{C_j}(K') < b_j.$$

These finitely many observation conditions define a neighbourhood of K contained in $\Box U$. For $K \cap U \neq \varnothing$, choose $x \in K \cap U$ and a basic interval $[a, b) \subseteq U$ containing x. Since $\mathscr{A}$ induces the topology on K, some $C \in \mathscr{A}$ satisfies $x \in K \cap C \subseteq [a, b)$. Requiring this fragment to be nonempty and imposing the same extremal bounds gives a neighbourhood contained in $\Diamond U$. Hence the observations generate the full Vietoris topology. Finite intersections of inverse images of countable local bases form a countable local base at each K. □

Lemma 4.6. *Let $X = D_{\mathrm{V}}$ and let m be a finite τ-smooth Borel measure on X. There is a countable family $C \subseteq \dot{O}(X)$ satisfying* (4.3).

Proof. Enumerate the observations in Lemma 4.5 as $\gamma_1, \gamma_2, \ldots$, and put

$$A_n = \{a \in \mathbb{R} : m(\gamma_n^{-1}(\{a\})) > 0\}.$$

Each A_n is at most countable, since these positive-mass fibres are disjoint and m is finite. Let τ_0 be generated, for all $n \geq 1$, by

$$\begin{aligned} &\gamma_n^{-1}((p,q)), && p,q \in \mathbb{Q},\ p<q,\\ &\gamma_n^{-1}([a,q)), && a \in A_n,\ q \in \mathbb{Q},\ a<q,\\ &\gamma_n^{-1}(\{*\}). \end{aligned}$$

There are only countably many generators, all open in X, so τ_0 is second-countable and coarser than the topology of X.

Take as a basis for the topology of X the finite intersections of sets of the form

$$\gamma_n^{-1}([a,b)) \qquad \text{and} \qquad \gamma_n^{-1}(\{*\}).$$

For a real interval condition, if $a \in A_n$, then $\gamma_n^{-1}([a,b))$ is τ_0-open, since it is the union of $\gamma_n^{-1}([a,q))$ over rational q with $a<q<b$. If $a \notin A_n$, replace it by $\gamma_n^{-1}((a,b))$. This set is τ_0-open because (a,b) is a union of intervals (p,q) with rational endpoints; the discarded set is contained in the null fibre $\gamma_n^{-1}(\{a\})$. Conditions at $*$ are already τ_0-open.

Let B be a finite intersection of these basic conditions and replace each interval condition as above. The resulting set B_0 is τ_0-open, satisfies $B_0 \subseteq B$, and has

$$B \setminus B_0 \subseteq \bigcup_{i=1}^{r} \gamma_{n_i}^{-1}(\{a_i\})$$

for finitely many null fibres. Hence $m(B_0) = m(B)$. Applying this construction to every member of a finite family of basic opens shows that each finite union F of basic opens contains a τ_0-open set F_0 such that $m(F_0) = m(F)$.

For $U \in \mathcal{O}(X)$, let $\mathcal{F}_U$ be the family of finite unions of basic opens contained in U. This is a directed family with union U. Each $F \in \mathcal{F}_U$ has a same-mass τ_0-open subset $F_0 \subseteq \operatorname{int}_{\tau_0} U$. Therefore $m(F) \leq m(\operatorname{int}_{\tau_0} U)$, and τ-smoothness gives

$$m(U) = \sup_{F \in \mathcal{F}_U} m(F) \leq m(\operatorname{int}_{\tau_0} U) \leq m(U).$$

Thus $m(U) = m(\operatorname{int}_{\tau_0} U)$.

Choose a countable basis $\mathcal{B}_0$ for τ_0, and let $\mathcal{C}$ consist of all finite unions of members of $\mathcal{B}_0$, including the empty union. The family $\mathcal{C}$ is countable, and every τ_0-open set is the directed union of the members of $\mathcal{C}$ that it contains. Applying τ-smoothness once more, together with the preceding equality, yields (4.3). □

Theorem 4.7. *The dcpo $D = \mathcal{Q}(\mathbb{R}_\ell)$ is a Fubini object.*

Proof. Let $\mu \in \mathbf{V}(D)$ be finite. By Lemma 4.4, there is a finite τ-smooth Borel measure m on $X = D_{\mathrm{V}}$ agreeing with μ on $\sigma(D)$. Put $\nu(U) = m(U)$ for $U \in \mathcal{O}(X)$. The identity $f : X \to \Sigma D$ is continuous and satisfies $f_*\nu = \mu$. By Lemmas 4.5 and 4.6, X and m satisfy the hypotheses of Proposition 4.2.

Fix a finite $\xi \in \mathbf{V}(\sigma(D))$. The inverse-image map

$$t : \sigma(D) \to \mathcal{O}(X), \qquad t(\mathcal{U}) = f^{-1}(\mathcal{U}),$$

preserves directed unions and is Scott-continuous. Define $\eta \in \mathbf{V}(\mathcal{O}(X))$ by

$$\eta(\mathcal{H}) = \xi(\{\mathcal{U} \in \sigma(D) : f^{-1}(\mathcal{U}) \in \mathcal{H}\}), \qquad \mathcal{H} \in \sigma(\mathcal{O}(X)).$$

This is the finite pushforward $t_*\xi$. Write

$$M_K^D = \{\mathcal{U} \in \sigma(D) : K \in \mathcal{U}\}, \qquad M_K^X = \{U \in \mathcal{O}(X) : K \in U\}.$$

Then $\eta(M_K^X) = \xi(M_K^D)$. The function $K \mapsto \xi(M_K^D)$ is Scott-continuous on D by Lemma 2.7; hence its pullback to X is lower semicontinuous. Using change of variables, Proposition 4.2, and the agreement of the measure and valuation integrals, we obtain

$$\begin{aligned}
\int_{\sigma(D)} \mu(\mathcal{U})\, d\xi(\mathcal{U}) &= \int_{\mathcal{O}(X)} \nu(U)\, d\eta(U) \\
&= \int_X \eta(M_K^X)\, dm(K) \\
&= \int_X \xi(M_K^D)\, d\nu(K) \\
&= \int_D \xi(M_K^D)\, d\mu(K).
\end{aligned}$$

By Corollary 2.9, the identity for all finite test valuations ξ implies that μ is central. Finally, Proposition 2.2 extends the conclusion from finite to all continuous valuations, so $\mathbf{V}(D) = \mathbf{Z}(D)$. □

5. Fubini descent through Scott-open set lattices

The following descent is an elementary application of canonical incidence. Combined with the known co-consonance results recalled below, it yields further Fubini objects.

Proposition 5.1. *If $\sigma(P)$ is a Fubini object, then so is P.*

Proof. Put $L = \sigma(P)$ and $M = \sigma(L)$. Fix $\mu \in \mathbf{V}(P)$ and $\xi \in \mathbf{V}(L)$, and define $\eta \in \mathbf{V}(M)$ by $\eta(\mathcal{W}) = \mu(\{x : M_x \in \mathcal{W}\})$, $\mathcal{W} \in \sigma(M)$. This is continuous because $n_P(x) = M_x$ is Scott-continuous by Lemma 2.7. Apply the Fubini property of L to ξ, η, and membership $E_L \subseteq L \times M$. Writing $N_U = \{\mathcal{H} \in M : U \in \mathcal{H}\}$ gives

$$\int_L \eta(N_U)\, d\xi(U) = \int_M \xi(\mathcal{H})\, d\eta(\mathcal{H}).$$

Since $n_P^{-1}(N_U) = U$, the left-hand side is $\int_L \mu(U)\, d\xi(U)$ and the right-hand side is $\int_P \xi(M_x)\, d\mu(x)$. Thus Proposition 2.8 gives centrality of every $\mu \in \mathbf{V}(P)$. □

For a space X, its open-set lattice carries the point topology τ_{p}, the Scott topology τ_{S} and the weak topology τ_{w} generated by $\{U \in \mathcal{O}(X) : U \not\subseteq V\}$, $V \in \mathcal{O}(X)$. The space is *co-consonant* when $\tau_{\mathrm{w}} = \tau_{\mathrm{S}}$ [6, Definition 7.1]. The weak and point topologies coincide: for closed $A \subseteq X$, the subbasic hitting set $\Diamond A = \{U : U \cap A \neq \varnothing\}$ equals $\bigcup_{x \in A} M_x$, while $M_x = \Diamond\overline{\{x\}}$. Thus co-consonance means that the point and Scott topologies on $\mathcal{O}(X)$ agree. In particular, $\mathbf{V}(X) = \mathbf{P}(X)$ follows directly from the definition of point-continuity.

Corollary 5.2. *If $\Sigma\sigma(P)$ is co-consonant, then P is a Fubini object.*

Proof. Co-consonance gives $\mathbf{V}(\sigma(P)) = \mathbf{P}(\sigma(P))$, so Corollary 2.5 and Proposition 5.1 apply. □

Theorem 5.3. *[6, Proposition 7.2] If X is countably based, then $\Sigma\mathcal{O}(X)$ is co-consonant.*

Corollary 5.4. *Every dcpo with second-countable Scott topology is a Fubini object.*

Proof. Apply Theorem 5.3 to $X = \Sigma P$, for which $\mathcal{O}(X) = \sigma(P)$, and use Corollary 5.2. □

Remark 5.5. A space X is *core-compact* if, for every $x \in U \in \mathcal{O}(X)$, some open neighbourhood V of x has the property that every directed open cover of U has a member containing V. Equivalently, $\mathcal{O}(X)$ is a continuous lattice [7, 8]. Here a lattice is *continuous* when every element is the directed supremum of elements way below it; $a \ll b$ means that every directed set whose supremum is at least b contains an element above a.

Centrality on core-compact Scott spaces is already known and is not an additional result here. Indeed, if ΣP is core-compact, apply [4, Theorem 4] to $X = \Sigma P$, to the dcpo $D = P$, and to the identity map $X \to P$. The resulting pushforward is the original valuation on P; Proposition 2.2 then supplies the extended convention used here. Moreover, when X is core-compact, $\mathcal{O}(X)$ is a continuous lattice and hence a continuous poset, so [15, Proposition 3.7] gives the co-consonance needed for the descent argument.

The second-countable corollary is not restricted to the core-compact case. Let $B = \omega^\omega$ be Baire space, where $\omega = \{0, 1, \ldots\}$ is discrete, and put

$$P = \sigma(\mathcal{O}(B)) = \mathcal{O}(\Sigma\mathcal{O}(B)).$$

The discussion immediately following [6, Proposition 6.16] shows that ΣP, namely the Scott space of $\mathcal{O}(\mathcal{O}(B))$, is quasi-Polish and not locally compact. In particular, it is countably based and sober. Since core-compactness and local compactness coincide on sober spaces [7, 8], ΣP is not core-compact, while Corollary 5.4 still applies.

Appendix A. Valuations on subspaces of countable Sorgenfrey products

Throughout this appendix, $S = \mathbb{R}_\ell$ and $S^{\mathbb{N}}$ carries the ordinary product topology. We prove that every continuous valuation on an arbitrary subspace $Y \subseteq S^{\mathbb{N}}$ has a central pushforward along every continuous map into the Scott space of a dcpo. The result follows from a source-incidence identity with no finiteness assumption on either valuation. No closedness or ambient Borel measurability of Y is required.

We first prove the identity for finite τ-smooth Borel measures, using discrete threshold minorants and a countable laminar family with a one-sided density property. This replaces the Scott–Fatou and Radon–Nikodym argument in Proposition 4.2. A Borel extension on spaces with a clopen basis and Heckmann's bounded approximation then give the result for arbitrary continuous valuations. The full Vietoris lift and the extremal observations are needed only for the application to D, which recovers Theorem 4.7. No identification of an ordinary product topology with a Scott topology is used.

A.1. The product representation and a measurable refinement

We record the representation needed for the final application before working with an arbitrary subspace of $S^{\mathbb{N}}$.

Proposition A.1. *There is a topological embedding $e : D_{\mathrm{V}} \to S^{\mathbb{N}}$. For every finite $\mu \in \mathbf{V}(D)$, its image $Y = e(D_{\mathrm{V}})$ carries a finite τ-smooth Borel measure m such that the continuous map $f : Y \to \Sigma D$, defined by $f(e(K)) = K$, satisfies $\mu(\mathcal{U}) = m(f^{-1}(\mathcal{U}))$ for every $\mathcal{U} \in \sigma(D)$.*

Proof. Enumerate the observations of Lemma 4.5 as $\gamma_1, \gamma_2, \ldots$. They generate the full Vietoris topology, which is Hausdorff by the embedding into the compact Hausdorff hyperspace used in Lemma 4.4. Consequently they separate points, and $K \mapsto (\gamma_n(K))_n$ is an embedding into $(S \sqcup \{*\})^{\mathbb{N}}$. The map

$$\iota : S \sqcup \{*\} \longrightarrow S^2, \qquad \iota(*) = (0,0), \qquad \iota(a) = (1,a),$$

is an embedding: its two summands have relatively clopen images, and the second projection identifies the nonisolated summand with S. Composing coordinatewise and reindexing the countably many coordinates gives e.

Let $\widetilde{m}$ be the measure on D_{V} supplied by Lemma 4.4. For Borel $B \subseteq Y$, put $m(B) = \widetilde{m}(e^{-1}(B))$. Since e is a homeomorphism onto Y, m is finite and τ-smooth. The map f is the composition of e^{-1} with the continuous identity $D_{\mathrm{V}} \to \Sigma D$, and the asserted equality follows from the defining property of $\widetilde{m}$. No ambient measurability of Y is required. □

The next elementary observation permits positive-mass coordinate fibres to be treated separately without a distributional transform.

Lemma A.2. *Let m be a finite τ-smooth Borel measure on a space (X, τ). If a topology τ' is obtained by adjoining countably many Borel sets and their complements as open sets, then τ and τ' have the same Borel sets and m is τ'-smooth. If (X, τ) is first-countable, then so is (X, τ').*

Proof. Let $\mathscr{B}$ be the countable Boolean algebra generated by the adjoined sets. Every τ'-open set is a countable union of sets $B \cap O$, where $B \in \mathscr{B}$ and $O \in \tau$: group all basic opens with the same Boolean part. Thus the two Borel σ-algebras agree.

To prove smoothness, let $(U_i)_i$ be directed in τ' and write $U_i = \bigcup_{B \in \mathscr{B}} (B \cap O_{i,B})$, with $O_{i,B} \in \tau$. Put $U = \bigcup_i U_i$ and $W_B = B \cap \bigcup_i O_{i,B}$. The family $(W_B)_{B \in \mathscr{B}}$ is countable and covers U. Given $\varepsilon > 0$, choose $B_1, \ldots, B_k$ such that

$$m\left(U \setminus \bigcup_{j=1}^{k} W_{B_j} \right) < \varepsilon/2;$$

the case $m(U) = 0$ is immediate. For each j, the finite Borel restriction of m to B_j is τ-smooth by (4.2). Applying it to the finite unions of the O_{i,B_j} gives a finite set I_j of indices with

$$m\left(W_{B_j} \setminus \bigcup_{i\in I_j}(B_j \cap O_{i,B_j})\right) < \varepsilon/(2k).$$

Directedness supplies an index i_0 such that U_{i_0} contains all U_i with $i \in \bigcup_j I_j$. Hence $m(U \setminus U_{i_0}) < \varepsilon$. This proves $m(U) = \sup_i m(U_i)$. Finally, intersect a countable original local base with the finite intersections of the countably many new subbasic sets containing the point. These intersections form a countable local base for τ'. □

Fix a subspace $Y \subseteq S^{\mathbb{N}}$ with a finite τ-smooth Borel measure m, and let $p_i : Y \to S$ be its coordinate maps. Define

$$A_i = \{a \in \mathbb{R} : m(p_i^{-1}(\{a\})) > 0\}, \qquad F_{i,a} = p_i^{-1}(\{a\}), \qquad N_i = Y \setminus \bigcup_{a\in A_i} F_{i,a}.$$

Each A_i is at most countable. Adjoin all sets $F_{i,a}$, $a \in A_i$, and N_i and their complements as opens, and denote the resulting space by Y'. By Lemma A.2, Y' is first-countable and carries the same finite τ-smooth Borel measure. The identity $q : Y' \to Y$ is continuous. After empty members are omitted, the family

$$\mathscr{P}_i = \{N_i\} \cup \{F_{i,a} : a \in A_i\}$$

is a countable clopen partition. On each of its parts, the coordinate p_i is either constant or has only null singleton fibres. The coordinate maps remain continuous into S.

Lemma A.3. *There is a countable family $C \subseteq \mathrm{O}(Y')$, closed under finite unions and containing $\varnothing$, such that*

$$m(U) = \sup\{m(C) : C \in C,\ C \subseteq U\} \qquad (U \in \mathrm{O}(Y')). \tag{A.1}$$

Proof. Let τ_0 be generated by all members of the partitions $\mathscr{P}_i$ and by the sets $p_i^{-1}((r, s))$, where $r < s$ are rational. This is a coarser second-countable topology on Y'.

A basis for the topology of Y' consists of finite intersections of partition conditions and sets $p_i^{-1}([a, b))$. For such an interval condition, put

$$W_i(a, b) = \begin{cases} p_i^{-1}((a, b)) \cup F_{i,a}, & a \in A_i, \\ p_i^{-1}((a, b)), & a \notin A_i. \end{cases}$$

The set $W_i(a, b)$ is τ_0-open and is contained in $p_i^{-1}([a, b))$. If $a \in A_i$, the two sets are equal. If $a \notin A_i$, their difference is the null fibre $p_i^{-1}(\{a\})$.

Replace every interval condition in a basic open set B by the corresponding $W_i(a, b)$, leaving the partition conditions unchanged. The resulting set B_0 is τ_0-open, satisfies $B_0 \subseteq B$, and differs from B by a subset of finitely many null fibres. Hence $m(B_0) = m(B)$. Taking unions shows that every finite union F of basic open sets contains a τ_0-open subset F_0 with $m(F_0) = m(F)$.

For $U \in \mathcal{O}(Y')$, let $\mathcal{F}_U$ be the directed family of finite unions of basic open sets contained in U. Its union is U. Every $F \in \mathcal{F}_U$ has a same-mass τ_0-open subset contained in $\operatorname{int}_{\tau_0} U$. By τ-smoothness,

$$m(U) = \sup_{F \in \mathcal{F}_U} m(F) \le m(\operatorname{int}_{\tau_0} U) \le m(U).$$

Thus $m(U) = m(\operatorname{int}_{\tau_0} U)$.

Choose a countable basis $\mathcal{B}_0$ for τ_0, and let C consist of all finite unions of members of $\mathcal{B}_0$, including the empty union. Every τ_0-open set is the directed union of the members of C that it contains. A final use of τ-smoothness yields (A.1). □

A.2. One-sided approximation

We first record the finite weighted consequence of Dilworth's theorem that is needed below. An antichain is a set of pairwise incomparable elements.

Lemma A.4. *Let V be a finite poset with weights $p_v \ge 0$, and put*

$$w = \max\left\{ \sum_{v \in H} p_v : H \subseteq V \text{ is an antichain} \right\}.$$

There are nonnegative coefficients t_C, indexed by the nonempty chains of V, such that

$$p_v = \sum_{C \ni v} t_C \quad (v \in V), \qquad \sum_C t_C = w. \tag{A.2}$$

Proof. Suppose first that all weights are rational. Choose an integer $d > 0$ such that every dp_v is integral. Replace each v by dp_v mutually incomparable copies, and declare a copy of v below a copy of v' precisely when $v < v'$ in V. Every antichain in the enlarged poset may be enlarged to contain all copies of every vertex that it meets. Its width is therefore dw. Dilworth's theorem [16, Theorem 1.1] partitions the enlarged poset into dw chains. Project the chains to V and divide their multiplicities by d. This gives (A.2).

For arbitrary real weights, choose nonnegative rational vectors $p^{(n)} = (p_v^{(n)})_{v \in V}$ converging to $p = (p_v)_{v \in V}$. Since V has only finitely many antichains, their maximum weights w_n converge to w. There are also only finitely many nonempty chains. The coefficient vectors supplied by the rational case are nonnegative and have total mass w_n, hence possess a convergent subsequence. Passing to the limit gives (A.2). □

We next isolate the measure-theoretic and combinatorial ingredients of the laminar construction. A compact set $H \subseteq \mathbb{R}^r$ is a *weak antichain* if it contains no two points x, z satisfying $x_i < z_i$ for every i.

Lemma A.5. *Let $r \ge 1$, and let β be a finite τ-smooth Borel measure on S^r such that all coordinate hyperplanes are β-null. Every Euclidean compact weak antichain $H \subseteq \mathbb{R}^r$ is β-null.*

Proof. For $x \in H$, put

$$Q_x = \prod_{i=1}^{r} [x_i, x_i + 1).$$

If $z \in H \cap Q_x$, then $z_i \geq x_i$ for every i. Since H is a weak antichain, equality holds in at least one coordinate. Hence

$$H \cap Q_x \subseteq \bigcup_{i=1}^{r} \{z : z_i = x_i\},$$

so $\beta(H \cap Q_x) = 0$. The Borel restriction $U \mapsto \beta(H \cap U)$ is a continuous valuation on S^r by (4.2). The finite unions of the sets Q_x, $x \in H$, form a directed family whose union contains H. Directed continuity therefore gives

$$\beta(H) = \sup_{F \subseteq H \text{ finite}} \beta\left(H \cap \bigcup_{x \in F} Q_x\right) = 0.$$

□

Let

$$R = \prod_{i=1}^{r} [a_i, b_i)$$

be a bounded half-open rectangle. A rectangular grid on R is obtained by choosing a finite partition of each interval $[a_i, b_i)$ into half-open subintervals and taking their products. Its cells are indexed by multi-indices. We order the cells by

$$R_v < R_w \quad \Longleftrightarrow \quad v_i < w_i \text{ for every } i.$$

Thus every point of R_v is strictly smaller in every coordinate than every point of R_w whenever $R_v < R_w$.

Lemma A.6. *Let β be as in Lemma A.5, and suppose that β is concentrated on a bounded half-open rectangle R. For any sequence of finite rectangular grids on R whose meshes tend to zero, the maximum β-weight of an antichain of cells tends to zero.*

Proof. Suppose otherwise. Passing to a subsequence, choose antichains $\mathcal{A}_n$ of cells whose unions have β-measure at least some $\varepsilon > 0$. Let

$$K_n = \bigcup_{C \in \mathcal{A}_n} \overline{C},$$

where the closures are Euclidean. The sets K_n are nonempty compact subsets of the fixed compact rectangle $\overline{R}$. The hyperspace of nonempty compact subsets of $\overline{R}$ is compact metrizable in the Hausdorff topology; see [13, Theorem 4.9.7]. After passing to a further subsequence, assume that K_n converges in Hausdorff distance to a nonempty compact set H.

The set H is a weak antichain. Indeed, if $x, z \in H$ and $x_i < z_i$ for every i, choose Euclidean neighbourhoods O_x, O_z whose coordinate projections are separated in this same order. For all sufficiently large n, the set K_n meets both neighbourhoods, and the mesh is smaller than the separation. The selected cells that meet O_x and O_z are then strictly ordered in every coordinate, contrary to the antichain property.

By Lemma A.5, $\beta(H) = 0$. On the other hand, $\beta(K_n) \geq \varepsilon$. Hausdorff convergence implies that for each $j \geq 1$, eventually $K_n \subseteq H^{1/j}$, the Euclidean $1/j$-neighbourhood of H. Hence $\beta(H^{1/j}) \geq \varepsilon$ for every j. Since $H^{1/j} \downarrow H$ and β is finite, continuity from above gives $\beta(H) \geq \varepsilon$, a contradiction. □

The next lemma supplies the exact clopen fragments used to realize the weights in Lemma A.4.

Lemma A.7. *Let $B \subseteq Y'$ be clopen, let $m(B) < \infty$, and suppose that for some coordinate i and real numbers $a < b$,*

$$B \subseteq p_i^{-1}([a,b)) \quad \text{and} \quad m(B \cap p_i^{-1}(\{t\})) = 0 \quad (t \in \mathbb{R}).$$

If $s_1, \dots, s_N \geq 0$ and $\sum_{j=1}^{N} s_j = m(B)$, then B has a clopen partition $B = B_1 \dot\cup \cdots \dot\cup B_N$ with $m(B_j) = s_j$ for every j.

Proof. Define

$$F(t) = m(B \cap p_i^{-1}([a,t))), \qquad a \leq t \leq b.$$

Then $F(a) = 0$ and $F(b) = m(B)$. The function F is continuous. For an increasing sequence $t_n \uparrow t$, this follows from continuity from below. For $t_n \downarrow t$, the sets $[a, t_n)$ decrease to $[a, t]$, and the difference between $[a, t]$ and $[a, t)$ is the m-null fibre at t.

Put $r_j = \sum_{h=1}^{j} s_h$. Since F is continuous and nondecreasing, its level sets can be chosen in increasing order. Thus there are

$$a = c_0 \leq c_1 \leq \cdots \leq c_N = b, \qquad F(c_j) = r_j.$$

Set

$$B_j = B \cap p_i^{-1}([c_{j-1}, c_j)).$$

The intervals $[c_{j-1}, c_j)$ are clopen in the Sorgenfrey line, so the sets B_j are clopen in Y'. They are pairwise disjoint, cover B, and satisfy $m(B_j) = s_j$. □

We now package the finite construction. Let I be a nonempty finite set of coordinates. A clopen set $B \subseteq Y'$ is called an I-macroblock if, for every $i \in I$, it is contained in $N_i \cap p_i^{-1}([a_i, b_i))$ for some bounded half-open interval $[a_i, b_i)$. Thus every $i \in I$ has null singleton fibres on B.

Lemma A.8. *Let $0 < \alpha < 1$, put $c = (1 - \alpha)/2$, and let B be a positive-mass I-macroblock. There are a finite clopen partition $\mathcal{R}_B$ of B, a finite family $\mathcal{T}_B$ of pairwise disjoint positive-mass clopen subsets of B, and a union G_B of members of $\mathcal{R}_B$ such that:*

1. *every $T \in \mathcal{T}_B$ has a representation*

$$T = B_{T,1} \dot\cup \cdots \dot\cup B_{T,\ell(T)},$$

   where the pieces belong to $\mathcal{R}_B$, have equal positive mass, and, whenever $j < k$, every point of $B_{T,j}$ is strictly smaller in all coordinates of I than every point of $B_{T,k}$;
2. *every positive-mass member of $\mathcal{R}_B$ belongs to exactly one of these representations;*
3. *$m(G_B) > c\, m(B)$;*
4. *if $y \in G_B \cap B_{T,j}$, then*

$$m\left(\bigcup_{k>j} B_{T,k}\right) > \alpha m(T),$$

   and every point of this later union is strictly to the right of y in all coordinates of I.

Proof. Let $\pi_I : B \to S^I$ be the active-coordinate map, and let β be the pushforward of the restricted measure $m|_B$. Explicitly, $\beta(A) = m(B \cap \pi_I^{-1}(A))$ for Borel $A \subseteq S^I$. Continuity of π_I and τ-smoothness of the Borel restriction $m|_B$ show that β is a finite τ-smooth Borel measure. It is concentrated on a bounded half-open rectangle, and every coordinate hyperplane is β-null. By Lemma A.6, choose a finite rectangular grid whose maximum weighted antichain has weight

$$w < c\, m(B).$$

For a grid cell R_v, put

$$B_v = B \cap \pi_I^{-1}(R_v), \qquad p_v = m(B_v).$$

Apply Lemma A.4 to the cell poset and the weights p_v, and omit all chains with $t_C = 0$.

Fix a cell with $p_v > 0$, enumerate the chains containing it as $C_1, \dots, C_N$, and choose any coordinate $i \in I$. The restriction of p_i to B_v has null singleton fibres and is contained in a bounded half-open interval. By Lemma A.7, there is a clopen partition

$$B_v = B_{v,C_1} \,\dot\cup\, \cdots \,\dot\cup\, B_{v,C_N}, \qquad m(B_{v,C_h}) = t_{C_h}.$$

Retain every zero-mass cell B_v as a separate member of the final partition. The resulting fragments, together with those zero-mass cells, form a finite clopen partition $\mathcal{R}_B$ of B.

For a chain $C = (v_1 < \cdots < v_\ell)$, set

$$T_C = \bigcup_{j=1}^{\ell} B_{v_j,C}.$$

The sets T_C are pairwise disjoint and clopen, and each has mass $\ell t_C > 0$. The order of the grid cells gives the asserted strict coordinate separation. Every positive-mass fragment belongs to exactly one T_C.

Declare $B_{v_j,C}$ to be good when $\ell - j > \alpha\ell$, and let G_B be the union of all good fragments. For a chain of length ℓ, the number of good indices is at least $(1-\alpha)\ell - 1$. Therefore

$$\begin{aligned} m(G_B) &\geq \sum_C ((1-\alpha)\ell(C) - 1)t_C \\ &= (1-\alpha)\sum_C \ell(C)t_C - \sum_C t_C \\ &= (1-\alpha)m(B) - w \\ &> c\, m(B). \end{aligned}$$

If $y \in B_{v_j,C}$ is good, the later fragments have total mass $(\ell - j)t_C > \alpha\ell t_C = \alpha m(T_C)$ and lie strictly to the right of y in every active coordinate. This proves all assertions. □

A family of sets is *laminar* if any two members are disjoint or one contains the other.

Lemma A.9. *Suppose that $m(Y') = 1$. For every $0 < \alpha < 1$, there are a countable laminar family $\mathcal{T}_\alpha$ of positive-mass clopen subsets of Y' and a Borel set E_α of measure one such that, simultaneously for every $U \in \mathcal{O}(Y')$,*

$$y \in E_\alpha \cap U \quad \Longrightarrow \quad \exists T \in \mathcal{T}_\alpha : \quad y \in T \ \text{ and } \ m(U \cap T) > \alpha m(T). \tag{A.3}$$

Proof. Put $c = (1-\alpha)/2$. For $n \geq 1$ and $i \leq n$, let

$$\mathscr{Q}_{i,n} = \{F_{i,a} : a \in A_i\} \cup \left\{N_i \cap p_i^{-1}([k2^{-n}, (k+1)2^{-n})) : k \in \mathbb{Z}\right\},$$

with empty sets omitted. This is a countable clopen partition of Y'. Begin with $\mathcal{T}_\alpha = \varnothing$. We construct recursively countable clopen partitions $\mathcal{P}_n$ of Y', together with open sets G_n.

Let $\mathcal{P}_0 = \{Y'\}$. Suppose that $\mathcal{P}_{n-1}$ has been constructed. For each $B \in \mathcal{P}_{n-1}$, take the common refinement of B by the finitely many partitions $\mathscr{Q}_{1,n}, \ldots, \mathscr{Q}_{n,n}$. Its nonempty members are called the stage-n macroblocks inside B. They form a countable clopen partition of B. For such a macroblock M, let

$$I(M) = \{i \leq n : M \subseteq N_i\}.$$

If $i \in I(M)$, then M is contained in a half-open interval of length 2^{-n} in coordinate i; if $i \notin I(M)$, that coordinate is constant on M.

If $m(M) = 0$, retain M as one part of $\mathcal{P}_n$ and put $G_M = \varnothing$. If $m(M) > 0$ and $I(M) = \varnothing$, retain M as one part, put M into $\mathcal{T}_\alpha$, and put $G_M = M$. If $m(M) > 0$ and $I(M) \neq \varnothing$, apply Lemma A.8 to M. Add its partition $\mathcal{R}_M$ to $\mathcal{P}_n$, add its chain sets $\mathcal{T}_M$ to $\mathcal{T}_\alpha$, and declare G_M to be its good set. Let G_n be the union of the good sets over all stage-n macroblocks. This construction makes $\mathcal{P}_n$ a countable clopen partition refining $\mathcal{P}_{n-1}$.

For every $B \in \mathcal{P}_{n-1}$, countable additivity over its stage-n macroblocks gives

$$m(G_n \cap B) \geq c\, m(B). \tag{A.4}$$

Every G_j, $j < n$, is a union of members of $\mathcal{P}_{n-1}$. Consequently, for integers $r \geq N$, put

$$A_{N,r} = \bigcap_{n=N}^{r} (Y' \setminus G_n).$$

The set $A_{N,r-1}$ is a union of members of $\mathcal{P}_{r-1}$, and (A.4) yields

$$m(A_{N,r}) \leq (1-c) m(A_{N,r-1}).$$

Induction gives

$$m(A_{N,r}) \leq (1-c)^{r-N+1}. \tag{A.5}$$

Letting $r \to \infty$ and then taking the countable union over N shows that

$$E_\alpha = \bigcap_{N \geq 1} \bigcup_{n \geq N} G_n$$

has measure one.

The family $\mathcal{T}_\alpha$ is countable. It is laminar because sets created at the same stage are disjoint. A set created at a later stage lies inside one member of every earlier partition. Each earlier chain set is a union of members of the partition created at its stage; therefore the later set is either contained in it or disjoint from it.

It remains to prove the common density property. Fix $y \in E_\alpha \cap U$. There is a basic neighbourhood V of y with $V \subseteq U$ of the form

$$V = \bigcap_{i \in F_0} P_i(y) \cap \bigcap_{i \in F_1} p_i^{-1}([y_i, y_i + \varepsilon_i)),$$

where F_0, F_1 are finite and $P_i(y)$ denotes the member of $\mathscr{P}_i$ containing y. Choose a stage n with $y \in G_n$ so large that

$$n \geq i \quad (i \in F_0 \cup F_1), \qquad 2^{-n} < \varepsilon_i \quad (i \in F_1).$$

Such stages exist because $y \in E_\alpha$; either family of requirements is vacuous when its index set is empty. Let M be the stage-n macroblock containing y.

If $I(M) = \varnothing$, then all coordinates occurring in V are constant on M, so $M \subseteq V$. The set $T = M$ belongs to $\mathcal{T}_\alpha$, and $m(U \cap T) = m(T) > \alpha m(T)$.

Suppose that $I(M) \neq \varnothing$. The point y lies in a good fragment $B_{T,j}$ of a chain set T. By Lemma A.8, the later fragments of T have total mass greater than $\alpha m(T)$. They satisfy all partition conditions defining V. In a coordinate $i \in F_1$ that is constant on M, they have value y_i. In an active coordinate, they lie strictly to the right of y_i and remain inside the same stage-n half-open interval as y_i; that interval has length $2^{-n} < \varepsilon_i$. Hence all later fragments lie in $V \subseteq U$, and

$$m(U \cap T) > \alpha m(T).$$

The set E_α was fixed before U, so the same full-measure set works simultaneously for every open U. □

A.3. Discrete tests and the incidence identity

A *discrete valuation* is a countable sum of Dirac valuations with finite nonnegative coefficients, interpreted as the directed supremum of its finite partial sums.

Lemma A.10. *Let L be a dcpo, let $\eta \in \mathbf{V}(L)$ be finite, and let $H_1 \supseteq \cdots \supseteq H_m \supseteq H_{m+1} = \varnothing$ be Scott-open sets. If each $H_j \setminus H_{j+1}$ has an at most countable order-coinitial subset, then there is a discrete valuation $\theta \leq \eta$ satisfying $\theta(H_j) = \eta(H_j)$ for $1 \leq j \leq m$.*

Proof. We first consider a finite continuous valuation ρ on a space Z and an at most countable set $A = \{a_1, a_2, \ldots\}$ such that $\rho(V) = \rho(Z)$ for every open neighbourhood V of A. Use an enumeration without repetition, finite when necessary, and put $A_n = \{a_1, \ldots, a_n\}$ and $A_0 = \varnothing$. By Lemma 3.1, $\sup_n \kappa_\rho(A_n) = \kappa_\rho(A) = \rho(Z)$. The capacity is monotone and submodular on finite sets:

$$\kappa_\rho(F \cup G) + \kappa_\rho(F \cap G) \leq \kappa_\rho(F) + \kappa_\rho(G).$$

Indeed, choose open neighbourhoods of F, G with masses arbitrarily close to their capacities, apply modularity, and use their union and intersection as neighbourhoods of $F \cup G, F \cap G$. All quantities are finite.

Set $b_n = \kappa_\rho(A_n) - \kappa_\rho(A_{n-1})$. These coefficients are nonnegative and sum to $\rho(Z)$. If $F = \{a_{i_1}, \ldots, a_{i_k}\}$, where $i_1 < \cdots < i_k$, put $F_j = \{a_{i_1}, \ldots, a_{i_j}\}$. Submodularity applied to $A_{i_j - 1}$ and F_j gives

$$b_{i_j} \leq \kappa_\rho(F_j) - \kappa_\rho(F_{j-1}).$$

Thus $\sum_{a_n \in F} b_n \le \kappa_\rho(F)$. For an open V, take finite subsets of $A \cap V$ to obtain $\sum_{a_n \in V} b_n \le \rho(V)$. Consequently $\theta_A = \sum_n b_n \delta_{a_n} \le \rho$ and $\theta_A(Z) = \rho(Z)$. When $A = \varnothing$, the hypothesis gives $\rho = 0$ and the empty sum suffices.

For the stated assertion, choose a countable order-coinitial $A_j \subseteq H_j \setminus H_{j+1}$ and define

$$\rho_j(V) = \eta(V \cap H_j) - \eta(V \cap H_{j+1}).$$

This is a continuous valuation. In particular, modularity rewrites it as

$$\rho_j(V) = \eta\big((V \cap H_j) \cup H_{j+1}\big) - \eta(H_{j+1}),$$

which proves monotonicity and preservation of directed unions; strictness and modularity follow from the first formula. Every Scott-open neighbourhood of A_j contains $H_j \setminus H_{j+1}$, so it has full ρ_j-mass. The preceding construction gives $\theta_j \le \rho_j$, supported on A_j, of total mass $\eta(H_j) - \eta(H_{j+1})$. Put $\theta = \sum_{j=1}^m \theta_j$. Since $\sum_j \rho_j(V) = \eta(V \cap H_1) \le \eta(V)$, we have $\theta \le \eta$. A point of A_j belongs to H_k exactly when $j \ge k$, and hence

$$\theta(H_k) = \sum_{j=k}^{m} \big(\eta(H_j) - \eta(H_{j+1})\big) = \eta(H_k).$$

□

Proposition A.11. *Let $Y \subseteq S^{\mathbb{N}}$ and let m be a finite τ-smooth Borel measure on Y. For every finite $\eta \in \mathbf{V}(\mathcal{O}(Y))$, the function $y \mapsto \eta(\{U \in \mathcal{O}(Y) : y \in U\})$ is lower semicontinuous and*

$$\int_{\mathcal{O}(Y)} m(U)\, d\eta(U) = \int_Y \eta(\{U \in \mathcal{O}(Y) : y \in U\})\, dm(y). \tag{A.6}$$

Proof. First note that for every first-countable space Z and continuous valuation ζ on $\mathcal{O}(Z)$, the function $g(z) = \zeta(M_z)$ is lower semicontinuous. Suppose otherwise. First-countability supplies $z_n \to z$ and a finite $r < g(z)$ such that $g(z_n) \le r$ for all n. The sets $K_N = \{z\} \cup \{z_n : n \ge N\}$ are compact, so

$$\mathcal{N}_N = \{U \in \mathcal{O}(Z) : K_N \subseteq U\}$$

is Scott open: a directed open cover of K_N has a finite subcover, and then one member covers it. The $\mathcal{N}_N$ increase to M_z, while $\mathcal{N}_N \subseteq M_{z_n}$ for every $n \ge N$. Thus $g(z) = \sup_N \zeta(\mathcal{N}_N) \le r$, a contradiction. The space Y is first-countable as a subspace of the countable product $S^{\mathbb{N}}$, and Y' is first-countable by Lemma A.2. Thus the argument applies to both spaces.

We establish the identity on Y' first. Write $\nu(U) = m(U)$, $L = \mathcal{O}(Y')$, and fix a finite $\zeta \in \mathbf{V}(L)$. The case of zero mass is immediate. Otherwise, by homogeneity in m, scale m to have total mass one. This scaling changes neither the refined topology nor the sets of null and positive coordinate fibres. For a finite $\rho \in \mathbf{V}(L)$, set

$$\Phi(\rho) = \int_{Y'} \rho(M_y)\, d\nu(y), \qquad \Psi(\rho) = \int_L \nu(U)\, d\rho(U).$$

These integrals are finite. For a finite discrete $\rho = \sum_i a_i \delta_{U_i}$, linearity and directed continuity give

$$\Phi(\rho) = \sum_i a_i \nu(U_i) = \Psi(\rho). \tag{A.7}$$

For $0 < r < 1$ let $H_r = \{U \in L : \nu(U) > r\}$. By (A.1), if $r < s < 1$, the set $\{C \in \mathcal{C} : r < \nu(C) \leq s\}$ is order-coinitial in $H_r \setminus H_s$: given U in that difference, choose $C \subseteq U$ with $\nu(C) > r$, and note that $\nu(C) \leq \nu(U) \leq s$. Similarly H_r itself has a countable order-coinitial subset. Apply Lemma A.10 to the finitely many thresholds occurring in

$$f_n(U) = 2^{-n} \sum_{j=1}^{2^n - 1} \mathbf{1}_{H_{j2^{-n}}}(U).$$

It gives a discrete $\theta_n \leq \zeta$ agreeing with ζ on these threshold opens. Since $f_n \leq \nu$,

$$\int_L f_n \, d\zeta = \int_L f_n \, d\theta_n \leq \Psi(\theta_n) = \Phi(\theta_n) \leq \Phi(\zeta).$$

The functions f_n are the strict lower dyadic approximants of $U \mapsto \nu(U) \in [0, 1]$; hence they increase pointwise to that function. Therefore $\Psi(\zeta) \leq \Phi(\zeta)$. No directedness of $(\theta_n)_n$ is asserted or needed.

For the other inequality, fix $0 < \alpha < 1$ and use Lemma A.9. For $T \in \mathcal{T}_\alpha$, put

$$\mathcal{E}_T = \{U \in L : \nu(U \cap T) > \alpha \nu(T)\}, \qquad R_\alpha = \bigcup_{T \in \mathcal{T}_\alpha} T \times \mathcal{E}_T.$$

Each $\mathcal{E}_T$ is Scott open, since intersection with the open set T preserves directed unions. Thus R_α is open in the ordinary product $Y' \times \Sigma L$. For every $y \in E_\alpha$, (A.3) gives $M_y \subseteq (R_\alpha)_y$.

For fixed $U \in L$, consider any finite family of selected sets T with $U \in \mathcal{E}_T$. Its inclusion-maximal members are pairwise disjoint and have the same union. Therefore

$$\alpha \nu\Big(\bigcup T\Big) \leq \sum_{T \text{ maximal}} \nu(U \cap T) \leq \nu(U).$$

The section R_α^U is the union of all selected members of $\mathcal{T}_\alpha$. Taking the directed union over their finite subfamilies and using continuity of ν yields $\nu(R_\alpha^U) \leq \alpha^{-1} \nu(U)$.

Apply the commutativity of the continuous-valuation monad, [9, Proposition 3.28], to the lower-semicontinuous characteristic function of the product-open set R_α. It gives the equality of the two iterated valuation integrals and, in particular, the lower semicontinuity of the section integrals. By [9, Proposition 4.3], these valuation integrals agree with the corresponding Lebesgue integrals with respect to m. The almost-everywhere section inclusion therefore gives

$$\Phi(\zeta) \leq \int_{Y'} \zeta((R_\alpha)_y) \, d\nu(y) = \int_L \nu(R_\alpha^U) \, d\zeta(U) \leq \alpha^{-1} \Psi(\zeta).$$

Letting $\alpha \uparrow 1$ proves $\Phi(\zeta) = \Psi(\zeta)$. The common set E_α is essential: it gives the inclusion of whole sections M_y, not merely an almost-everywhere assertion for each separately chosen open U.

Finally, return to Y and the given test valuation η. The inverse-image map $t : \mathcal{O}(Y) \to \mathcal{O}(Y')$, $t(U) = q^{-1}(U)$, is Scott-continuous. Define $\zeta = t_*\eta$, explicitly $\zeta(\mathcal{H}) = \eta(\{U : q^{-1}(U) \in \mathcal{H}\})$. Then

$$\zeta(M_y^{Y'}) = \eta(M_{q(y)}^{Y}), \qquad m(q^{-1}(U)) = m(U).$$

The two topologies have the same Borel sets and q is the identity on the underlying set, so $q_*m = m$. Applying the identity on Y' and using change of variables twice gives

$$\begin{aligned}\int_{\mathcal{O}(Y)} m(U)\, d\eta(U) &= \int_{\mathcal{O}(Y')} m(V)\, d\zeta(V) \\ &= \int_{Y'} \zeta(M_y^{Y'})\, dm(y) \\ &= \int_{Y} \eta(M_y^{Y})\, dm(y).\end{aligned}$$

Here the measure and valuation integrals of the lower-semicontinuous section functions agree by [9, Proposition 4.3]. This is (A.6). □

A.4. Arbitrary continuous valuations and central pushforwards

The preceding identity is stated for finite Borel measures. To apply it to every continuous valuation, we first give the elementary extension argument needed here.

Lemma A.12. *Let X be a space with a clopen basis. Every finite continuous valuation ν on X extends to a finite τ-smooth Borel measure m on X.*

Proof. For $A \subseteq X$, define

$$m^*(A) = \inf\{\nu(V) : A \subseteq V \in \mathcal{O}(X)\}.$$

This function is finite, monotone, and zero at the empty set. Finite subadditivity and directed continuity of ν imply

$$\nu\left(\bigcup_{n\geq 1} V_n\right) \leq \sum_{n\geq 1} \nu(V_n) \qquad (V_n \in \mathcal{O}(X)).$$

Given $A_n \subseteq X$ and $\varepsilon > 0$, choose open $V_n \supseteq A_n$ with $\nu(V_n) < m^*(A_n) + \varepsilon 2^{-n}$. Then

$$m^*\left(\bigcup_{n\geq 1} A_n\right) \leq \nu\left(\bigcup_{n\geq 1} V_n\right) \leq \sum_{n\geq 1} m^*(A_n) + \varepsilon.$$

Letting ε tend to zero proves countable subadditivity, so m^* is an outer measure. Monotonicity of ν gives $m^*(U) = \nu(U)$ for every open U.

We check the Carathéodory criterion for an open set U. Fix $A \subseteq X$, an open $V \supseteq A$, and $\varepsilon > 0$. The clopen subsets of U form a directed family with union U; hence some clopen $C \subseteq U$ satisfies $\nu(V \cap U) \leq \nu(V \cap C) + \varepsilon$. Since $V \setminus C$ is open and contains $A \setminus U$, modularity gives

$$\begin{aligned}m^*(A \cap U) + m^*(A \setminus U) &\leq \nu(V \cap U) + \nu(V \setminus C) \\ &\leq \nu(V \cap C) + \nu(V \setminus C) + \varepsilon \\ &= \nu(V) + \varepsilon.\end{aligned}$$

Taking the infimum over V and then letting ε tend to zero gives one inequality in the Carathéodory criterion; the other is subadditivity of m^*. Thus every open set is m^*-measurable. The Carathéodory theorem, in the form of [17, Theorem 3.2], now shows that the restriction of m^* to the Borel σ-algebra is a measure m. It is finite and agrees with ν on open sets. Consequently its restriction to opens preserves directed unions, so m is τ-smooth. □

Theorem A.13. *Let $Y \subseteq S^{\mathbb{N}}$ and put $M_y = \{U \in \mathcal{O}(Y) : y \in U\}$. For every $\nu \in \mathbf{V}(Y)$ and every $\eta \in \mathbf{V}(\mathcal{O}(Y))$, the function $y \mapsto \eta(M_y)$ is lower semicontinuous and*

$$\int_{\mathcal{O}(Y)} \nu(U)\, d\eta(U) = \int_Y \eta(M_y)\, d\nu(y). \tag{A.8}$$

Neither valuation is required to have finite mass, and the integrals may take the value $+\infty$.

Proof. The space Y is first-countable. The convergent-sequence argument at the beginning of the proof of Proposition A.11 shows that $y \mapsto \eta(M_y)$ is lower semicontinuous for arbitrary η; that argument imposes no finiteness assumption.

Suppose first that ν and η are finite. Half-open intervals form a clopen basis for S, so finite-coordinate cylinders and their traces give a clopen basis for Y. By Lemma A.12, ν extends to a finite τ-smooth Borel measure m on Y. Apply Proposition A.11 to m and η. Since $m(U) = \nu(U)$ on open sets and their integrals agree on nonnegative lower-semicontinuous functions by [9, Proposition 4.3], this gives (A.8).

For arbitrary ν and η, the spaces Y and $\Sigma\mathcal{O}(Y)$ are T_0. By Theorem 2.1, write $\nu = \sup_i \nu_i$ and $\eta = \sup_j \eta_j$ as directed suprema of finite continuous valuations. For each y, the set M_y is Scott open in $\mathcal{O}(Y)$, so $\eta(M_y) = \sup_j \eta_j(M_y)$. Directed continuity of integration in the integrand and the valuation gives

$$\begin{aligned}\int_{\mathcal{O}(Y)} \nu(U)\, d\eta(U) &= \sup_{i,j} \int_{\mathcal{O}(Y)} \nu_i(U)\, d\eta_j(U) \\ &= \sup_{i,j} \int_Y \eta_j(M_y)\, d\nu_i(y) \\ &= \int_Y \eta(M_y)\, d\nu(y).\end{aligned}$$

All suprema here are directed, and no subtraction of infinite values occurs. This proves the theorem. □

Corollary A.14. *Let $Y \subseteq S^{\mathbb{N}}$, let P be a dcpo, and let $f : Y \to \Sigma P$ be continuous. For every $\nu \in \mathbf{V}(Y)$, the valuation $\mu \in \mathbf{V}(P)$ defined by*

$$\mu(U) = \nu(f^{-1}(U)), \qquad U \in \sigma(P),$$

is central.

Proof. Fix a finite $\xi \in \mathbf{V}(\sigma(P))$. The inverse-image map $t : \sigma(P) \to \mathcal{O}(Y)$, $t(U) = f^{-1}(U)$, preserves unions and is Scott-continuous. Define the finite continuous valuation η on $\mathcal{O}(Y)$ by

$$\eta(\mathcal{H}) = \xi(\{U \in \sigma(P) : f^{-1}(U) \in \mathcal{H}\}), \qquad \mathcal{H} \in \sigma(\mathcal{O}(Y)).$$

For $M_x^P = \{U \in \sigma(P) : x \in U\}$ and $M_y^Y = \{V \in \mathcal{O}(Y) : y \in V\}$, we have $t^{-1}(M_y^Y) = M_{f(y)}^P$, hence $\eta(M_y^Y) = \xi(M_{f(y)}^P)$. By Theorem A.13 and change of variables,

$$\begin{aligned}\int_{\sigma(P)} \mu(U)\, d\xi(U) &= \int_{\mathcal{O}(Y)} \nu(V)\, d\eta(V) \\ &= \int_Y \eta(M_y^Y)\, d\nu(y) \\ &= \int_Y \xi(M_{f(y)}^P)\, d\nu(y) \\ &= \int_P \xi(M_x^P)\, d\mu(x).\end{aligned}$$

The last change of variables is valid because $x \mapsto \xi(M_x^P)$ is Scott-continuous by Lemma 2.7 and directed continuity of ξ. The finite-test criterion, Corollary 2.9, therefore proves that μ is central, without a finiteness assumption on ν or μ. □

Here Y retains its ordinary product-subspace topology. The conclusion concerns central valuations on target dcpos; it does not identify Y with a Scott space or call Y itself a Fubini object in the sense of Definition 2.6. In particular, it is stronger than an identity restricted to lower-semicontinuous functions on an ordinary topological product: for any dcpo Q, any $\lambda \in \mathbf{V}(Q)$, and any Scott-continuous $h : P \times Q \to \overline{\mathbb{R}}_+$, it gives

$$\int_Y \left(\int_Q h(f(y), z)\, d\lambda(z) \right) d\nu(y) = \int_Q \left(\int_Y h(f(y), z)\, d\nu(y) \right) d\lambda(z).$$

This is (2.2) for the central valuation μ, followed by change of variables along f. Neither the masses nor the integrand need be bounded.

Remark A.15. The generality of Theorem A.13 can also be obtained from the main-text method. Indeed, for a finite τ-smooth Borel measure m on $Y \subseteq S^{\mathbb{N}}$, let p_i be the coordinate maps and put $A_i = \{a : m(p_i^{-1}(\{a\})) > 0\}$. Each A_i is countable. Let τ_0 be generated by

$$p_i^{-1}((r, s)), \quad r, s \in \mathbb{Q},\ r < s, \qquad p_i^{-1}([a, r)), \quad a \in A_i,\ r \in \mathbb{Q},\ a < r.$$

This is a coarser second-countable topology on Y. For a basic half-open cylinder, each interval condition with left endpoint in A_i is already τ_0-open; replace every other condition $p_i \in [a, b)$ by $p_i \in (a, b)$. The loss is contained in finitely many null coordinate fibres. Finite unions of basic cylinders therefore have same-mass τ_0-open subsets. Taking their directed union inside an arbitrary open U gives $m(U) = m(\mathrm{int}_{\tau_0} U)$. Finite unions from a countable basis of τ_0 provide the inner approximation required by Proposition 4.2. That proposition proves the finite-measure identity, and the extension and bounded approximation above give Theorem A.13.

This alternative derivation is not used in the product-subspace proof. The distinction lies in the proof of the incidence identity, not in the class of subspaces covered: the main text uses Scott–Fatou and Radon–Nikodym, whereas the proof above uses discrete threshold minorants and the laminar density construction on a common set of full measure in Lemma A.9.

A.5. Application to the Sorgenfrey–Smyth dcpo

Alternative proof of Theorem 4.7. Let $\mu \in \mathbf{V}(D)$ be finite. By Proposition A.1, choose $Y \subseteq S^{\mathbb{N}}$, a finite τ-smooth Borel measure m on Y, and a continuous $f : Y \to \Sigma D$ with $\mu(\mathcal{U}) = m(f^{-1}(\mathcal{U}))$ for every $\mathcal{U} \in \sigma(D)$. The restriction $\nu(U) = m(U)$, $U \in \mathcal{O}(Y)$, is a continuous valuation, and Corollary A.14 shows that μ is central. Finally, Proposition 2.2 extends the conclusion from finite to all continuous valuations on D. □

Declaration on the use of artificial intelligence

LLMs were used during the preparation of this manuscript to assist in generating and developing mathematical proofs, proof checking, and language editing. In particular, the product-subspace proof method and its extension to arbitrary continuous valuations presented in Appendix A were completely generated by LLMs. The author takes full responsibility for the mathematical arguments, references, and final content of the manuscript.

References

[1] C. Jones and G. D. Plotkin, *A probabilistic powerdomain of evaluations*, in Proceedings of the Fourth Annual Symposium on Logic in Computer Science, IEEE Computer Society, 1989, pp. 186–195, doi:10.1109/LICS.1989.39173.

[2] C. Jones, *Probabilistic Non-Determinism*, Ph.D. thesis, University of Edinburgh, 1990, Technical Report ECS-LFCS-90-105.

[3] X. Jia, B. Lindenhovius, M. Mislove, and V. Zamdzhiev, *Commutative monads for probabilistic programming languages*, in 36th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2021), IEEE, 2021, pp. 1–14, doi:10.1109/LICS52264.2021.9470611. Extended version: arXiv:2102.00510; Theorem 21 is cited from the extended version.

[4] X. Jia, M. Mislove, and V. Zamdzhiev, *The central valuations monad*, in 9th Conference on Algebra and Coalgebra in Computer Science (CALCO 2021), Leibniz International Proceedings in Informatics, vol. 211, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2021, Article 18, pp. 18:1–18:5 (Early Ideas), doi:10.4230/LIPIcs.CALCO.2021.18.

[5] J. Goubault-Larrecq and X. Jia, *Separating minimal valuations, point-continuous valuations, and continuous valuations*, Mathematical Structures in Computer Science 31 (2021), 614–632, doi:10.1017/S0960129521000384. Numbered results are cited from arXiv:2109.00426v1.

[6] M. de Brecht and T. Kawai, *On the commutativity of the powerspace constructions*, Logical Methods in Computer Science 15 (2019), no. 3, Paper No. 13, 25 pp., doi:10.23638/LMCS-15(3:13)2019.

[7] G. Gierz, K. H. Hofmann, K. Keimel, J. D. Lawson, M. Mislove, and D. S. Scott, *Continuous Lattices and Domains*, Encyclopedia of Mathematics and its Applications, vol. 93, Cambridge University Press, Cambridge, 2003.

[8] J. Goubault-Larrecq, *Non-Hausdorff Topology and Domain Theory*, New Mathematical Monographs, vol. 22, Cambridge University Press, Cambridge, 2013.

[9] T. Fritz, P. Perrone, and S. Rezagholi, *Probability, valuations, hyperspace: three monads on Top and the support as a morphism*, Mathematical Structures in Computer Science 31 (2021), no. 8, 850–897, doi:10.1017/S0960129521000414. Preprint version: arXiv:1910.03752v3; all numbered results refer to this version.

[10] R. Heckmann, *Spaces of valuations*, Annals of the New York Academy of Sciences 806 (1996), 174–200, doi:10.1111/j.1749-6632.1996.tb49168.x.

[11] X. Xu and Z. Yang, *Coincidence of the upper Vietoris topology and the Scott topology*, Topology and its Applications 288 (2021), Article 107480, doi:10.1016/j.topol.2020.107480. Numbered results are cited from arXiv:2003.06542v1.

[12] D. H. Fremlin, *Measure Theory. Volume 2: Broad Foundations*, Torres Fremlin, Colchester, 2001. Online edition, Corollary 232G and Remark 232H(e).

[13] E. Michael, *Topologies on spaces of subsets*, Transactions of the American Mathematical Society 71 (1951), no. 1, 152–182, doi:10.1090/S0002-9947-1951-0042109-4.

[14] M. Álvarez-Manilla, A. Jung, and K. Keimel, *The probabilistic powerdomain for stably compact spaces*, Theoretical Computer Science 328 (2004), no. 3, 221–244, doi:10.1016/j.tcs.2004.06.021.

[15] Z. He and B. Zhao, *Some problems about consonant and co-consonant spaces*, Acta Mathematica Sinica, English Series 41 (2025), no. 4, 1152–1164, doi:10.1007/s10114-025-3080-2. Proposition 3.7 is cited from arXiv:2211.10202v2, entitled *Some problems about co-consonance of topological spaces*.

[16] R. P. Dilworth, *A decomposition theorem for partially ordered sets*, Annals of Mathematics (2) 51 (1950), no. 1, 161–166, doi:10.2307/1969503.

[17] K. Keimel and J. D. Lawson, *Measure extension theorems for T_0-spaces*, Topology and its Applications 149 (2005), nos. 1–3, 57–83, doi:10.1016/j.topol.2004.02.019. Theorem 3.2 is cited from the author version dated 7 October 2010, https://www2.mathematik.tu-darmstadt.de/~logik/keimel/Papers/keimelawson20.pdf.